L5a: Multiple Asset Geometric Brownian Motion

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Objectives for Today

Today we extend geometric Brownian motion from one asset to many correlated assets, estimate the covariance that couples them from data, and introduce a way to sample portfolio weights.

Today's concepts

- Model correlated assets: the multiple asset GBM equation with a shared shock vector and a covariance factor $\mathbf{A}$ with $\mathbf{A}\mathbf{A}^{\top} = \mathbf{C}$, and its exact one-step transition.
- Estimate the covariance: center the growth-rate data matrix, compute the sample covariance and correlation, convert to the GBM covariance rate with the time step, and check the structural properties.
- Describe and sample portfolio weights: buy-and-hold wealth on the long-only simplex, the Dirichlet distribution and its moments, and why sampling explores but does not optimize.

Lectures and Examples

Lecture:
CHEME-5660-L5a-Lecture-MultipleAsset-GBM-Fall-2026.ipynb

Example 1:
CHEME-5660-L5a-Example-OOS-SAGBM-Fall-2026.ipynb

Example 2:
CHEME-5660-L5a-Example-CovarianceMatrix-Fall-2026.ipynb

Example 3:
CHEME-5660-L5a-Example-Dirichlet-PortfolioWeights-Fall-2026.ipynb

The first example predicts prices a single asset GBM model never saw and measures how often they stay inside its bands. The second computes the covariance matrix for our dataset and converts it to the GBM covariance rate. The third samples portfolio weights on the simplex.

Concept Review: Single Asset GBM Model

Let $S(t) > 0$ be the share price. With arithmetic drift μ (inverse years), volatility $\sigma > 0$ (inverse years to the one-half power), and Wiener increment $dW(t)$, single asset GBM is:

$$\frac{dS(t)}{S(t)} = \mu\, dt + \sigma\, dW(t).$$

- Drift versus growth: the mean growth rate is $\bar{g} = \mu - \sigma^2/2$, the expected one-step growth rate; μ and $\bar{g}$ share units and differ by the half-variance correction.
- Risk: σ sets the growth-rate standard deviation $\sigma/\sqrt{\Delta t}$; from data, $\hat{\sigma} = \sigma_g\sqrt{\Delta t}$.
- Constancy: both are assumed constant; the out-of-sample example shows what that costs.

Concept Review: The One-Step Transition and Out of Sample

On a grid $t_j = j\Delta t$ of N steps the price advances by the exact one-step transition of L4b, with independent standard normal shocks Z_j:

$$\boxed{S_{t_j} = S_{t_{j-1}} \exp\!\Big[\bar{g}\,\Delta t + \sigma\sqrt{\Delta t}\,Z_j\Big], \qquad j = 1, \dots, N.}$$

For one fixed horizon T the same solution reads $S_T = S_0 \exp[\bar{g}T + \sigma\sqrt{T}\,Z]$; the one-step form is what simulates paths.

In L4b we estimated $\bar{g}$ and σ from data the model had already seen. Example 1 asks the model about prices it has not seen: parameters from 2014 to 2024, predictions for all of 2025, and a count of how often the observed price stays inside the model's bands.

Example 1: CHEME-5660-L5a-Example-OOS-SAGBM-Fall-2026.ipynb

Catch-up: The GBM Trade Rule

If we did not reach the trade-rule probability at the end of L4b, we pick it up now. Buy n_0 shares at S_0, sell at S_T after a holding period $T > 0$, with benchmark growth rate g_b; the scaled NPV is $\rho_T = \text{NPV}(g_b, T)/(n_0 S_0) = (S_T/S_0)e^{-g_b T} - 1$. For a target $\rho_\star > -1$, with Φ the standard normal cumulative distribution function, single asset GBM gives the probability of clearing it in closed form:

$$\boxed{\mathbb{P}(\rho_T > \rho_\star) = 1 - \Phi\left(\frac{\ln(1+\rho_\star) + g_b T - \bar{g}\,T}{\sigma\sqrt{T}}\right).}$$

The derivation is the GBM Trade Rule section of the L4b lecture, and the L4b trade-rule example evaluates it for a firm in our dataset. Everything else today extends the price model itself, from one asset to many.

Multiple Asset Geometric Brownian Motion (MAGBM) Model

Independently simulated assets never move together, and co-movement is the point of a portfolio. Consider assets $\mathcal{P} = \{1, \dots, M\}$ with share prices $S_i(t)$ and drifts μ_i, a covariance rate $\mathbf{C} \in \mathbb{R}^{M\times M}$ (inverse years, $C_{ii} = \sigma_i^2$), a loading matrix $\mathbf{A}$ with $\mathbf{A}\mathbf{A}^\top = \mathbf{C}$, and M independent Wiener processes $W_\ell(t)$:

$$\frac{dS_i(t)}{S_i(t)} = \underbrace{\mu_i\,dt}_{\text{drift}} + \underbrace{\sum_{\ell=1}^{M} A_{i\ell}\,dW_\ell(t)}_{\text{correlated noise}}, \qquad i \in \mathcal{P}.$$

Itô's lemma on $\ln S_i$ gives each asset its own half-variance correction: the mean growth rate is $\bar{g}_i = \mu_i - C_{ii}/2$, exactly the single asset case with $\sigma_i^2 = C_{ii}$.

Indices from here on: i, j assets, ℓ shock coordinates, k time periods.

Why the $\mathbf{A}\mathbf{A}^\top$ Factorization?

A factorization $\mathbf{A}\mathbf{A}^\top = \mathbf{C}$ is a Cholesky decomposition when $\mathbf{A}$ is lower triangular; the symmetric square root $\mathbf{C}^{1/2}$ is another valid choice. Any covariance factor with $\mathbf{A}\mathbf{A}^\top = \mathbf{C}$ will do.

- Independent noise becomes correlated noise: the W_ℓ are independent, but multiplying their increments by $\mathbf{A}$ mixes them into the co-movement we observe.
- It reproduces the covariance rate: over dt, the noise terms of assets i and j have covariance:

$$\mathrm{Cov}\Big(\sum_\ell A_{i\ell}\, dW_\ell,\ \sum_\ell A_{j\ell}\, dW_\ell\Big) = \sum_\ell A_{i\ell}A_{j\ell}\, dt = (\mathbf{A}\mathbf{A}^\top)_{ij}\, dt = C_{ij}\, dt.$$

So $\mathbf{C}$ is the covariance per unit time, and the factorization is what makes the simulated assets have the covariance we asked for.

The Exact One-Step Transition

Let $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_M)$, with $\mathbf{I}_M$ the $M \times M$ identity, be a vector of independent standard normal shocks, drawn fresh at every step and shared by all assets within that step. Over a step Δt:

$$\boxed{S_i(t+\Delta t) = S_i(t)\exp\Big[\underbrace{\bar{g}_i\,\Delta t}_{\text{mean growth}} + \underbrace{\sqrt{\Delta t}\,(\mathbf{AZ})_i}_{\text{correlated shock}}\Big], \qquad i \in \mathcal{P},}$$

with $(\mathbf{AZ})_i = \sum_\ell A_{i\ell} Z_\ell$.

- Exact at the grid points for constant parameters; not a time-stepping approximation.
- Redraw $\mathbf{Z}$ at every step: a path is built from independent increments. One fixed horizon T can instead be sampled directly with a single shock vector scaled by $\sqrt{T}$.

The Scaling Rule

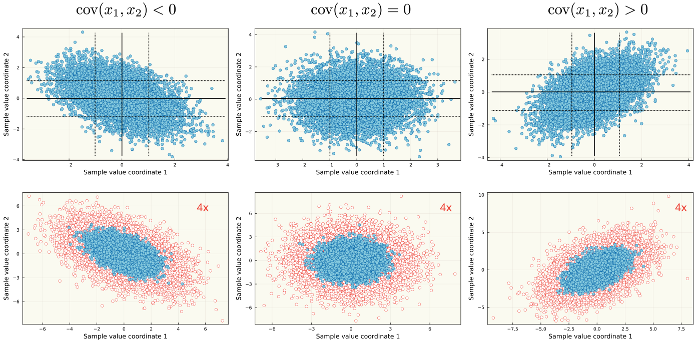
Divide the transition by $S_i(t)$ and take logarithms: the one-step log return is $r_i = \ln(S_i(t+\Delta t)/S_i(t))$ and the growth rate is $g_i = r_i/\Delta t$. Collect them in vectors $\mathbf{r}$ and $\mathbf{g} = \mathbf{r}/\Delta t$, the drifts in μ, and the mean growth rates in $\bar{\mathbf{g}}$:

$$\mathbf{g} = \bar{\mathbf{g}} + \frac{1}{\sqrt{\Delta t}}\mathbf{A}\mathbf{Z} \implies \mathbb{E}[\mathbf{g}] = \bar{\mathbf{g}} = \mu - \tfrac{1}{2}\,\mathrm{diag}(\mathbf{C}), \quad \underbrace{\mathrm{Cov}(\mathbf{g})}_{\text{growth-rate}} = \frac{\mathbf{C}}{\Delta t}, \quad \underbrace{\mathrm{Cov}(\mathbf{r})}_{\text{log-return}} = \mathbf{C}\,\Delta t.$$

Three different matrices, related by powers of Δt: growth-rate covariance (inverse years squared), log-return covariance (dimensionless), and the covariance rate $\mathbf{C}$ (inverse years). Keeping them apart is the multi-asset version of L4b's $\hat{\sigma} = \sigma_g\sqrt{\Delta t}$.

The model needs $\mathbf{C}$, and $\mathbf{C}$ comes from data.

Covariance: Direction and Scale



Top row: samples with negative, zero, and positive covariance. Bottom row: the same clouds with the covariance multiplied by four; the spread doubles, the direction of co-movement does not change.

Empirical Covariance Matrix

Take $N \geq 2$ equally spaced, aligned periods of growth rates for the M firms in $\mathcal{P}$; $g_k^{(i)}$ is firm i's growth rate in period k, and $g_i' = \frac{1}{N}\sum_k g_k^{(i)}$ its sample mean. The empirical growth-rate covariance $\hat{\Sigma}_g \in \mathbb{R}^{M\times M}$ has entries:

$$\hat{\Sigma}_{g,ij} = \frac{1}{N-1}\sum_{k=1}^{N} \underbrace{\left(g_k^{(i)} - g_i'\right)}_{\text{deviation}}\left(g_k^{(j)} - g_j'\right), \qquad i, j \in \mathcal{P}.$$

The diagonal $\hat{\Sigma}_{g,ii}$ is firm i's growth-rate variance, so $\sqrt{\hat{\Sigma}_{g,ii}}$ is L3a's σ_g for that firm (inverse years), not the volatility. The sign of an off-diagonal entry is the direction of the relationship; its magnitude depends on both firms' scales.

Correlation

The standardized strength of a relationship is the correlation ρ_{ij} of firms i and j, defined whenever both variances are positive:

$$\boxed{\rho_{ij} = \frac{\hat{\Sigma}_{g,ij}}{\sqrt{\hat{\Sigma}_{g,ii}\,\hat{\Sigma}_{g,jj}}} \in [-1, 1] \quad \Longleftrightarrow \quad \hat{\Sigma}_{g,ij} = \rho_{ij}\sqrt{\hat{\Sigma}_{g,ii}\,\hat{\Sigma}_{g,jj}}.}$$

- Correlation is what to quote when comparing pairs; covariance is what the model needs.
- Zero correlation means no linear relationship in this sample; it does not imply independence.
- Correlations are the same for the growth-rate covariance, the log-return covariance, and the covariance rate: the scaling cancels in the ratio.

The Data Matrix and the Sample Covariance

Arrange the growth rates in a data matrix $\mathbf{G} \in \mathbb{R}^{N \times M}$ (rows are periods, columns are firms) and let $\mathbf{g}' = (g'_1, \dots, g'_M)^\top$ be the vector of sample means. With $\mathbf{1} \in \mathbb{R}^N$ the vector of ones, the centered matrix and the covariance are:

$$\tilde{\mathbf{G}} = \mathbf{G} - \mathbf{1}\,\mathbf{g}'^\top, \qquad \boxed{\hat{\Sigma}_g = \frac{1}{N-1}\tilde{\mathbf{G}}^\top\tilde{\mathbf{G}}.}$$

$\mathbf{1}\,\mathbf{g}'^\top$ is an outer product: an $N \times M$ matrix whose every row is the vector of sample means, so subtracting it centers each column.

One matrix product replaces $M(M+1)/2$ pairwise sums.

From the Growth-Rate Covariance to the GBM Covariance Rate

The scaling rule says $\mathrm{Cov}(\mathbf{g}) = \mathbf{C}/\Delta t$, so the estimate of the covariance rate the model needs is:

$$\boxed{\widehat{\mathbf{C}} = \Delta t\,\widehat{\Sigma}_g.}$$

- Its diagonal returns the L4b volatilities firm by firm: $\hat{\sigma}_i = \sqrt{\widehat{C}_{ii}} = \sqrt{\Delta t}\sqrt{\widehat{\Sigma}_{g,ii}}$.
- The log-return covariance is $\widehat{\Sigma}_r = \Delta t^2\,\widehat{\Sigma}_g = \Delta t\,\widehat{\mathbf{C}}$.
- Daily data, $\Delta t = 1/252$: multiplying a daily log-return covariance by 252 estimates $\widehat{\mathbf{C}}$; it does not estimate $\widehat{\Sigma}_g$.

Covariance Matrix Properties

Every covariance estimate must satisfy these, and they are worth checking numerically:

- Elements: diagonal entries are variances (non-negative); off-diagonal entries are covariances whose sign gives the direction.
- Symmetry: $\hat{\Sigma}_{g,ij} = \hat{\Sigma}_{g,ji}$, directly from the definition; $\hat{\mathbf{C}} = \Delta t\,\hat{\Sigma}_g$ inherits it.
- Positive semidefinite: for any weight vector $\mathbf{v} \in \mathbb{R}^M$ the quadratic form is a squared length:

$$\mathbf{v}^\top \hat{\Sigma}_g \mathbf{v} = \frac{1}{N-1}\|\tilde{\mathbf{G}}\mathbf{v}\|_2^2 \geq 0,$$

  so a portfolio variance can never come out negative, and the L5b optimization rests on this.

Caveats Before Calling the Estimate Simulation-Ready

Only semidefinite. Duplicated assets, an exact linear dependence, or too few observations make $\hat{\Sigma}_g$ singular; since $\tilde{\mathbf{G}}$ has rank at most $N-1$, it is always singular once $M \geq N$. An unpivoted Cholesky factorization then fails; an eigenvalue or singular-value square root still works. Diagonal "jitter" restores invertibility but changes the model and must be documented.

Sampling error. $M(M+1)/2$ distinct entries from N observations are noisy when M is not small relative to N; pairwise estimates shift across windows and correlations change in stressed markets. Shrinkage and factor models trade flexibility for stability.

Example 2: CHEME-5660-L5a-Example-CovarianceMatrix-Fall-2026.ipynb

Portfolio Weights and Dirichlet Sampling

Let $W_0 > 0$ be initial wealth and $\mathbf{w} = (w_1, \dots, w_M)^\top$ the initial dollar fractions placed in each asset. Fully invested and long only: $w_i \geq 0$, $\sum_i w_i = 1$, so $\mathbf{w}$ lies on the simplex Δ^{M-1} (a segment for two assets, a triangle for three).

Buying $n_i = w_i W_0 / S_i(0)$ fractional shares and holding them, wealth at time t is:

$$\boxed{W_t = \sum_{i \in \mathcal{P}} n_i S_i(t) = W_0 \sum_{i \in \mathcal{P}} w_i \frac{S_i(t)}{S_i(0)}.}$$

The weights drift: the fraction in asset i at time t is $w_i(t) = n_i S_i(t)/W_t$. A constant-weight portfolio that trades back to $\mathbf{w}$ is a different, rebalanced strategy with its own turnover and costs.

Dirichlet Sampling of the Simplex

Let $\alpha = (\alpha_1, \dots, \alpha_M)$ with concentration parameters $\alpha_i > 0$ and $\alpha_0 = \sum_i \alpha_i$. A random vector $\mathbf{W} \sim \mathrm{Dirichlet}(\alpha)$ lies in the interior of Δ^{M-1} (positive weights summing to one) with:

$$\mathbb{E}[W_i] = \frac{\alpha_i}{\alpha_0}, \qquad \mathrm{Var}(W_i) = \frac{\alpha_i(\alpha_0 - \alpha_i)}{\alpha_0^2(\alpha_0 + 1)}, \qquad \mathrm{Cov}(W_i, W_j) = -\frac{\alpha_i \alpha_j}{\alpha_0^2(\alpha_0 + 1)} \; (i \neq j).$$

- Covariances are negative: the weights compete for one unit of wealth.
- Every $\alpha_i = 1$: constant density with respect to volume on the simplex, not uniform in each component.
- Symmetric $\alpha_i = \alpha$: below one favors allocations near the boundary relative to uniform; large α clusters near equal weights $1/M$; unequal α_i tilt toward the assets with larger concentration.

Sampling Is Not Optimization

A Dirichlet draw is a candidate weight vector; it becomes good or bad only when an objective is evaluated for it. The simplest: to first order the one-step growth rate of a portfolio held at weights $\mathbf{w}$ is $g_p = \mathbf{w}^\top \mathbf{g}$ (a one-period, fixed-weight statistic), whose estimated variance is:

$$\underbrace{\mathbf{w}^\top \hat{\Sigma}_g \mathbf{w}}_{\text{portfolio growth-rate variance}} \geq 0.$$

- Sampling explores the simplex; the concentration vector is the dial that says where.
- L5b optimizes: it turns the estimated means and covariance into the minimum-variance portfolio; its portfolio example simulates buy-and-hold wealth with correlated MAGBM paths for a chosen $\mathbf{w}$.

Example 3: CHEME-5660-L5a-Example-Dirichlet-PortfolioWeights-Fall-2026.ipynb

Optional Advanced Material

Advanced 1: advanced/covariance-estimation/
CHEME-5660-L5a-Advanced-CovarianceEstimation-Fall-2026.ipynb

Advanced 2: advanced/rolling-correlation/
CHEME-5660-L5a-Advanced-RollingCorrelation-Fall-2026.ipynb

Optional, not prerequisites for L5b. The first measures how noisy a large sample covariance is (its eigenvalues against the Marchenko–Pastur law), shrinks it toward a structured target, and shows what that does to a minimum-variance portfolio; the second tracks pairwise correlations through 2014 to 2024 with rolling and exponentially weighted windows and watches them spike in stressed markets.

Notebooks: lectures/week-5/L5a/advanced, also linked from the lecture notebook.

Summary

Today we extended GBM to many correlated assets, estimated the covariance that couples them from historical growth rates, and introduced Dirichlet sampling as a way to explore portfolio weights.

Key takeaways

- Correlated assets need a shared shock and a matrix square root: one shock vector per step, mixed by $\mathbf{A}$ with $\mathbf{A}\mathbf{A}^{\top} = \mathbf{C}$.
- Covariance is estimated from centered growth rates and scaled by the time step: symmetric and positive semidefinite by construction, $\hat{\mathbf{C}} = \Delta t\,\hat{\Sigma}_g$; singularity and sampling error must be checked.
- Weights live on the simplex, and Dirichlet draws explore it: the concentration vector says where the draws land; sampling is not optimization.

Next, we turn the estimated means and covariance into the minimum-variance portfolio and the efficient frontier.

Today: one shock vector, many assets, and the matrix that ties them together.